

Name Teacher



MORIAH COLLEGE

Year 12 2007 Pre-Trial

Extension 1 MATHEMATICS

Time Allowed: 2 hours plus 5 minutes reading time

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Instructions:

- Answer every question. All questions are of equal value.
- Show all necessary working. Draw clear, well labelled diagrams.

(a) Find the exact value of $\cos\left(\frac{5\pi}{4}\right)$. **1**

(b) Find:

(i) $\int e^{3x-5} dx$ **1**

(ii) $\frac{d}{dx}(\tan^{-1} 4x)$. **2**

(c) Evaluate $\int_0^1 \frac{1}{\sqrt{2-x^2}} dx$. **2**

(d) The point P divides the line AB externally in the ratio $1:4$. Find P if A is $(2, 1)$ and B is $(-4, 5)$. **2**

(e) $\lim_{x \rightarrow 0} \frac{\sin x}{3x}$ **1**

(f) Find the values of x which satisfy $\frac{3}{x-1} \leq 3$. **3**

- (a) (i) Show that the curves $y = 4x$ and $y = x^3$ intersect at the point where $x = 2$. **2**

- (ii) Find the acute angle between the two curves at $x = 2$, correct to the nearest degree. **2**

- (b) Find the derivative of $f(x) = 5x^2 - x$ from first principles using the definition : **2**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} .$$

- (c) Find the exact value of $\int_0^{\frac{\pi}{3}} 3 \cos x \sin^2 x \, dx$. **2**

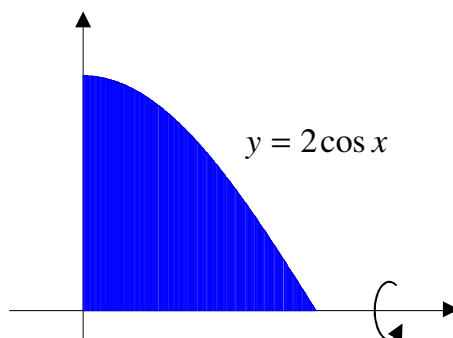
- (d) Find the exact values of

(i) $\sin^{-1}\left(\cos \frac{\pi}{6}\right)$. **1**

(ii) $\cos\left(2 \sin^{-1} \frac{3}{7}\right)$. **3**

(a) (i) Show that $\cos^2 x = \frac{1}{2} \cos 2x + \frac{1}{2}$. 2

- (ii) The shaded region bounded by the curve $y = 2 \cos x$ and the coordinate axes is rotated around the x -axis to form a solid.

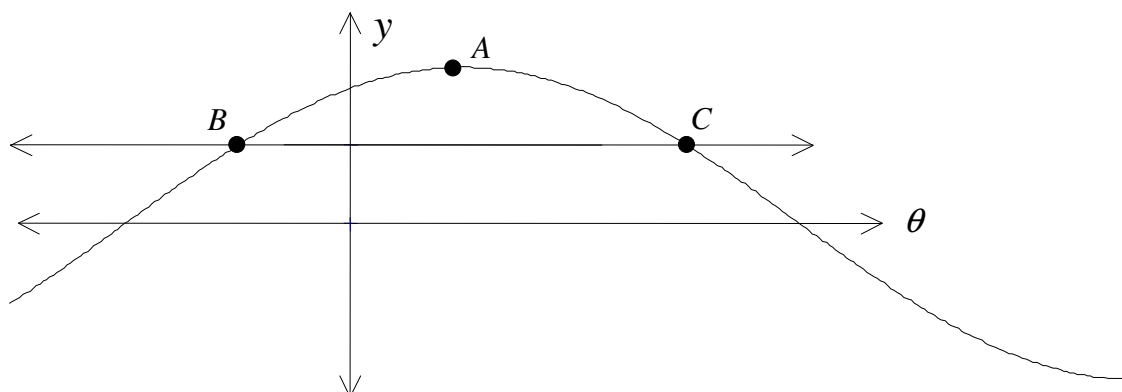


Using part (i), find the volume of the solid in terms of π . 3

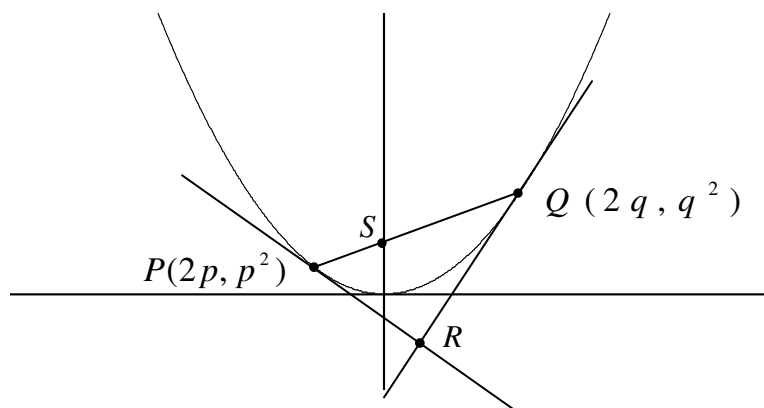
(b) (i) Express $\sin 2\theta + \sqrt{3} \cos 2\theta$ in the form $R \sin(2\theta + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$ and $R > 0$. 2

- (ii) The graphs of $y = \sin 2\theta + \sqrt{3} \cos 2\theta$ and $y = 1$ are shown on the real number plane below. The point A is a relative maximum and the points B and C are points of intersection.

Find the coordinates of the points A, B and C. 5



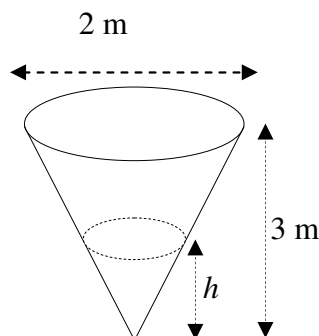
(a)



The diagram shows the graph of the parabola $x^2 = 4y$. The line PQ is a focal chord which intersects the y -axis at S .

- (i) Prove that the equation of the tangent to the parabola at P is $y = px - p^2$. 2
- (ii) The tangents at P and Q meet at point R . Find the coordinates of R . 2
- (iii) Find the equation of the line PQ and hence, show that $pq = -1$. 2
- (iv) Find the Cartesian equation of the locus of R , as P moves on the parabola. 2
- (b) Given that $\frac{d}{dx}(x \log x) = \log x + 1$, find the exact value of $\int_e^{e^2} \frac{1 + \log x}{x \log x} dx$. 2
- (c) Use the substitution $t = \tan \theta$ to show that $\sqrt{\frac{1 - \sin 2\theta}{1 + \sin 2\theta}} = \frac{1 - \tan \theta}{1 + \tan \theta}$, 2
where $-1 < \tan \theta \leq 1$.

(a)



A conical container, with diameter 2 metres and height 3 metres, is being filled with water at a rate of $0.5 \text{ m}^3/\text{min}$. The volume of water V in the container at any time t , is given by:

$$V = \frac{\pi h^3}{27} \text{ (cubic metres)}$$

- (i) Find $\frac{dV}{dh}$ and hence, show that the rate of increase in the height h , of the water at any time t is given by: 2

$$\frac{dh}{dt} = \frac{9}{2\pi h^2} \text{ (metres/min)}$$

- (ii) Find the exact rate of increase in the height h , of the liquid when the container is $\frac{1}{4}$ full. Give your answer in simplest form. 2

- (b) The line $y = mx$ is a tangent to the curve $y = \log x$.

- (i) Find the value of m . 3
- (ii) Hence, find the range of values of k such that the equation $kx = \log x$ has two distinct roots. 2

- (c) Prove that $\frac{\sin 5\theta}{\sin \theta} - \frac{\cos 5\theta}{\cos \theta} = 4\cos 2\theta$ 3

- (a) Use the principle of mathematical induction to prove that for all integers $n \geq 1$, **3**

$$2 + 10 + 24 + \dots + n(3n - 1) = n^2(n + 1).$$

- (b) Consider the function $f(x) = \frac{x}{\sqrt{1-x^2}}$

- (i) Write down the domain of $f(x)$. **1**

- (ii) Show that $f'(x) = \frac{1}{(1-x^2)^{\frac{3}{2}}}$. **2**

- (iii) Find the equation of the tangent to $f(x) = \frac{x}{\sqrt{1-x^2}}$ at $x = 0$. **1**

- (iv) Sketch a graph of $y = f(x)$ clearly showing the tangent at $x = 0$, and any asymptotes. **2**

- (v) Sketch an accurate graph of $y = f^{-1}(x)$ on the same set of axes. **1**

- (vi) Find the inverse function $y = f^{-1}(x)$. **2**

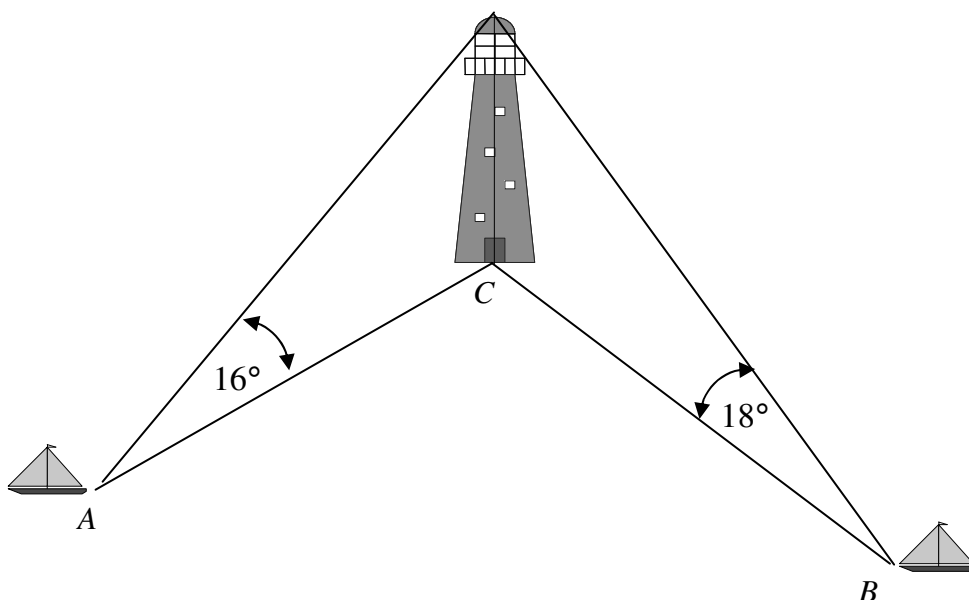
- (a) A boat sails from a point A to a point B .

At point A the captain of the ship measures the angle of elevation of the top of a lighthouse as 16° and the bearing of the lighthouse as 040° .

At point B the captain of the ship measures the angle of elevation of the top of the lighthouse as 18° and the bearing of the lighthouse as 340° .

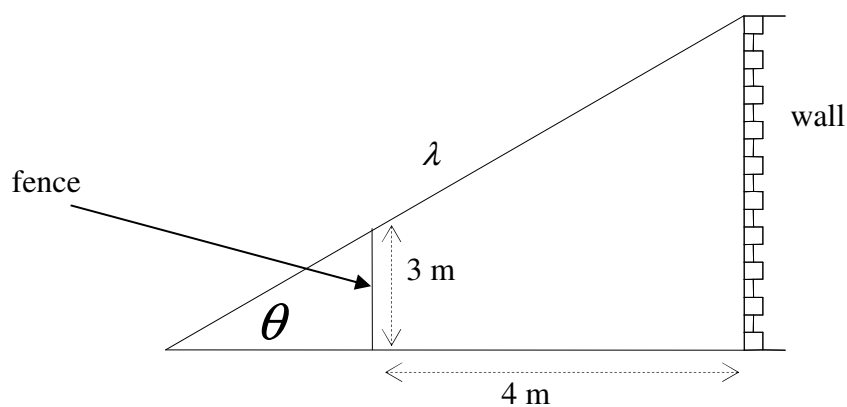
The top of the lighthouse is known to be 80 m above sea level.

The diagram below shows the angles of elevation of the top of the lighthouse from A and B .



- | | |
|--|----------|
| (i) Draw a bearing diagram showing the relative positions of A , B and C and use your diagram to explain why $\angle ACB = 60^\circ$. | 1 |
| (ii) Hence, find the distance between A and B , correct to the nearest metre. | 3 |
| (iii) Hence, find the bearing of B from A , to the nearest degree. | 2 |

- (b) A ladder λ m long is leaning against a vertical wall so that it just touches the top of a fence that is 3 m high and 4 m from the wall. The ladder is inclined at an angle of θ radians to the ground.

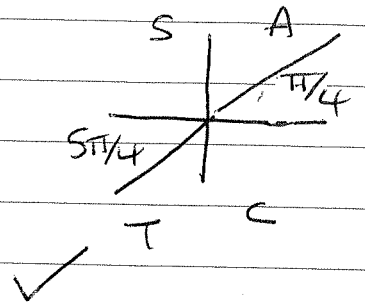


- (i) Prove that the length of the ladder $\lambda = \frac{3}{\sin \theta} + \frac{4}{\cos \theta}$. 2
- (ii) Show that if $\lambda = 10$ then the angle θ satisfies the equation 2
 $\sin(2\theta) = \sin(\theta + \varphi)$, where $\tan \varphi = \frac{3}{4}$.
- (iii) Hence, find the value(s) of θ . 2

Question 1:

$$(a) \cos\left(\frac{5\pi}{4}\right) = -\cos\frac{\pi}{4}$$

$$= -\frac{1}{\sqrt{2}}$$



$$(b) (i) \int e^{3x-5} dx$$

$$= \frac{1}{3} e^{3x-5} + C$$

$$(ii) \frac{d}{dx} (\tan^{-1} 4x)$$

$$= \frac{1}{1+(4x)^2} \cdot 4$$

$$= \frac{4}{1+16x^2}$$

$$(c) \int_0^1 \frac{1}{\sqrt{2-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{(\sqrt{2})^2 - x^2}} dx$$

$$= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1$$

$$= \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0$$

$$= \frac{\pi}{4} - 0$$

$$= \frac{\pi}{4}$$

(d) A (2, 1) B (-4, 5)

$\swarrow \quad \searrow$
 $-1 : 4$

$$P \left(\frac{2 \times 4 + -1 \times -4}{3}, \frac{4 \times 1 + -1 \times 5}{3} \right) \checkmark$$

$$= \left(4, -\frac{1}{3} \right) \checkmark$$

(e) $\lim_{x \rightarrow 0} \frac{\sin x}{3x}$ ✓

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= \frac{1}{3} \times 1$$

$$= \frac{1}{3} \checkmark$$

(f) $\frac{3}{x-1} \leq 3$

(i) $x \neq 1$ ✓



(ii) Assume true

$$\frac{3}{x-1} = 3 \checkmark$$

$$3x - 3 = 3$$

$$3x = 6$$

$$x = 2$$

(iii) test $x = 0$

$$-3 \leq 3 \quad (\checkmark \text{ True})$$

$$\therefore x < 1, x \geq 2 \checkmark$$

$$\begin{aligned}
 2) a) i) \quad x^3 &= 4x \\
 x^3 - 4x &= 0 \\
 x(x^2 - 4) &= 0 \\
 x=0, \quad x^2 &= 4 \\
 x &= \pm 2
 \end{aligned}$$

$\therefore$ the curves $y=4x$ and $y=x^3$ intersect at the point where $x=2$.

$$\begin{aligned}
 ii) \quad \frac{dy}{dx} &= 4 & \frac{dy}{dx} &= 3x^2 \\
 & & \text{at } x=2 & \\
 & & \frac{dy}{dx} &= 3 \times 2^2 \\
 & & &= 3 \times 4 \\
 & & &= 12
 \end{aligned}$$

$$\begin{aligned}
 \tan \theta &= \left| \frac{4-12}{1+(4)(12)} \right| \\
 &= \left| \frac{-8}{1+48} \right| \\
 &= \frac{8}{49}
 \end{aligned}$$

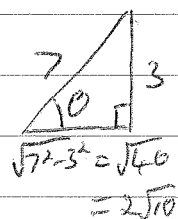
$$\begin{aligned}
 \theta &= \text{INV tan} \left(\frac{8}{49} \right) \\
 &= 9^\circ
 \end{aligned}$$

$$\begin{aligned}
 b) \quad f'(x) &= \lim_{h \rightarrow 0} \frac{5(x+h)^2 - (x+h) - (5x^2 - x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{5(x^2 + 2xh + h^2) - x - h - 5x^2 + x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - x - h - 5x^2 + x}{h} \\
 &= \lim_{h \rightarrow 0} 10x + 5h - 1 \\
 &= 10x - 1
 \end{aligned}$$

$$\begin{aligned}
 c) \quad \int_0^{\frac{\pi}{3}} 3 \cos x \sin^2 x \, dx &= \left[\sin^3 x \right]_0^{\frac{\pi}{3}} \\
 &= \sin^3 \frac{\pi}{3} - \sin^3 0 \\
 &= \left(\frac{\sqrt{3}}{2} \right)^3 - 0 \\
 &= \frac{\sqrt{27}}{8} \\
 &= \frac{3\sqrt{3}}{8}
 \end{aligned}
 \quad \frac{d}{dx} \sin^3 x = 3 \cos x \sin^2 x$$

$$\begin{aligned}
 d) \quad i) \quad \sin^{-1} \left(\cos \frac{\pi}{6} \right) &= \sin^{-1} \left(\sin \left(\frac{\pi}{2} - \frac{\pi}{6} \right) \right) \\
 &= \sin^{-1} \left(\sin \left(\frac{\pi}{3} \right) \right) \\
 &= \frac{\pi}{3}
 \end{aligned}$$

$$\begin{aligned}
 ii) \quad \cos \left(\sin^{-1} \frac{3}{7} + \sin^{-1} \frac{3}{7} \right) &= \cos^2 \left(\sin^{-1} \frac{3}{7} \right) - \sin^2 \left(\sin^{-1} \frac{3}{7} \right) \\
 &= \left(\cos \left(\sin^{-1} \frac{2\sqrt{10}}{7} \right) \right)^2 - \left(\frac{3}{7} \right)^2 \\
 &= \left(\frac{2\sqrt{10}}{7} \right)^2 - \frac{9}{49} \\
 &= \frac{40}{49} - \frac{9}{49} = \frac{31}{49}
 \end{aligned}$$



Question 3:

$$(a) (i) \cos 2x = \cos^2 x - \sin^2 x$$
$$\cos 2x = \cos^2 x - (1 - \cos^2 x)$$
$$\cos 2x = \cos^2 x - 1 + \cos^2 x.$$

$(-\frac{1}{2}$
for
setting
out)

$$\cos 2x = 2\cos^2 x - 1$$
$$2\cos^2 x = \cos 2x + 1$$
$$\cos^2 x = \frac{1}{2}(\cos 2x + 1)$$

$$\cos^2 x = \frac{1}{2} \cos 2x + \frac{1}{2} \quad \checkmark$$

$$(ii) V = \pi \int_0^{\pi/2} y^2 dx \quad y = 2\cos x$$

$$= \pi \int_0^{\pi/2} (2\cos x)^2 dx \quad \checkmark$$

$y=0:$
 $2\cos x = 0$
 $\cos x = 0$
 $x = \pi/2.$

$$= \pi \int_0^{\pi/2} (4\cos^2 x) dx$$

$$= 4\pi \int_0^{\pi/2} \frac{1}{2} (\cos 2x + 1) dx.$$

$$= 2\pi \int_0^{\pi/2} (\cos 2x + 1) dx \quad \checkmark$$

$$= 2\pi \left[\frac{\sin 2x}{2} + x \right]_0^{\pi/2}.$$

$$= 2\pi \left[\frac{\pi}{2} - 0 \right]$$

$$= \pi^2 \text{ units}^3 \quad \checkmark$$

$$(b) (i) \sin 2\theta + \sqrt{3} \cos 2\theta = R \sin(2\theta + \alpha)$$

$$R = \sqrt{1+3} = 2.$$

✓

$$\sin(2\theta + \alpha) = \sin 2\theta \cos \alpha + \sin \alpha \cos 2\theta$$

$$\cos \alpha = 1$$

$$\tan \alpha = \sqrt{3}$$

$$\sin \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

✓

$$R \sin(2\theta + \alpha) = 2 \sin(2\theta + \frac{\pi}{3})$$

$$(ii) \sin 2\theta + \sqrt{3} \cos 2\theta = 2 \sin(2\theta + \frac{\pi}{3})$$

For A, $y=2$.

$$\Rightarrow 2 \sin(2\theta + \frac{\pi}{3}) = 2$$

$$\sin(2\theta + \frac{\pi}{3}) = 1$$

$$\sin^{-1}(1) = 2\theta + \frac{\pi}{3}$$

$$\frac{\pi}{2} = 2\theta + \frac{\pi}{3}$$

$$2\theta = \frac{\pi}{2} - \frac{\pi}{3}$$

$$2\theta = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{12}$$

✓

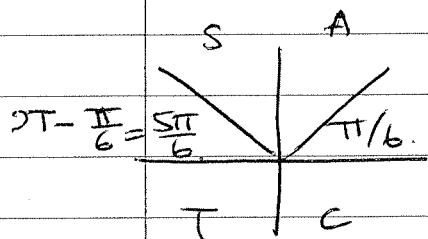
$$\therefore A \left(\frac{\pi}{12}, 2 \right)$$

For B and C: $y=1$.

$$2 \sin\left(2\theta + \frac{\pi}{3}\right) = 1 \quad \checkmark$$

$$\sin\left(2\theta + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$\sin^{-1}\left(\frac{1}{2}\right) = 2\theta + \frac{\pi}{3}$$



$$\frac{\pi}{6} = 2\theta + \frac{\pi}{3}, \quad \frac{5\pi}{6} = 2\theta + \frac{\pi}{3}$$

$$2\theta = \frac{\pi}{6} - \frac{\pi}{3}, \quad 2\theta = \frac{5\pi}{6} - \frac{\pi}{3}$$

$$2\theta = -\frac{\pi}{6}$$

$$2\theta = \frac{\pi}{2}$$

$$\theta = -\frac{\pi}{12}$$

$$\theta = \frac{\pi}{4}$$

$$\therefore B = \left(-\frac{\pi}{12}, 1\right) \quad \checkmark$$

$$C = \left(\frac{\pi}{4}, 1\right) \quad \checkmark$$

4) a) i) $x^2 = 4y$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{2x}{4}$$

$$m_{\text{tangent}} = \frac{2x}{4}$$

$$= \frac{x}{2}$$

at P, $x = 2p$

$\therefore$ eqⁿ of tangent:

$$\Rightarrow m_{\text{tangent}} = \frac{2p}{2}$$

$$= p \quad \textcircled{1}$$

$$y - p^2 = p(x - 2p)$$

$$y = px - 2p^2 + p^2$$

$$\underline{y = px - p^2} \quad \textcircled{1}$$

ii) tangent at Q must have eqⁿ: $y = qx - q^2$

$\therefore$ at intersection of tangent from P and tangent from Q:

$$px - p^2 = qx - q^2$$

$$(p - q)x = p^2 - q^2$$

$$x = \frac{p^2 - q^2}{(p - q)}$$

$$x = \frac{(p + q)(p - q)}{(p - q)}$$

$$x = p + q \quad \textcircled{1}$$

Sub x into $y = px - p^2$

$$y = p(p + q) - p^2$$

$$= p^2 + pq - p^2$$

$$= pq$$

$$\therefore R = (p + q, pq) \quad \textcircled{1}$$

$$\text{iii)} \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{p^2 - q^2}{2p - 2q}$$

$$= \frac{(p+q)(p-q)}{2(p-q)}$$

$$= \frac{p+q}{2}$$

PQ passes through $P(2p, p^2)$

$$\Rightarrow y - p^2 = \frac{p+q}{2}(x - 2p)$$

$$y - p^2 = \frac{(p+q)x}{2} - p^2 - pq \quad \checkmark \textcircled{1}$$

$$y = \frac{(p+q)x}{2} - pq$$

PQ passes through $S(0, 1)$:

$$\Rightarrow 1 = -pq$$

$$pq = -1 \quad \checkmark \textcircled{1}$$

$$\text{iv)} \Rightarrow R = (p+q, -1)$$

$\therefore R$ moves on the line $y = -1$ $\checkmark \checkmark$

$$b) \frac{d}{dx} (x \log x) = \log x + 1$$

$$\begin{aligned} \int_e^{e^2} \frac{1 + \log x}{x \log x} dx &= \left[\log |x \log x| \right]_e^{e^2} \quad \checkmark (1) \\ &= \log |e^2 \log e^2| - \log |e \log e| \\ &= \log |2e^2 \log e| - \log |e| \\ &= \log \left| \frac{2e^2}{e} \right| \quad \text{Ans} \\ &= \log |2e| \\ &= \log 2 + 1 \quad \checkmark (1) \end{aligned}$$

$$\begin{aligned} c) \text{ LHS} &= \sqrt{\frac{1 - \frac{2t}{1+t^2}}{1 + \frac{2t}{1+t^2}}} & \text{RHS} &= \frac{1 - \tan \theta}{1 + \tan \theta} \\ &= \sqrt{\frac{1+t^2-2t}{1+t^2+2t}} \\ &= \sqrt{\frac{(t-1)^2}{(t+1)^2}} \\ &= \frac{t-1}{t+1} \\ &= \frac{1 - \tan \theta}{\tan \theta + 1} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$5) a) i) \frac{dV}{dh} = \frac{3\pi h^2}{27}$$

$$= \frac{\pi h^2}{9}$$

①

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{9}{\pi h^2} \times \frac{1}{2}$$

$$= \frac{9}{2\pi h^2}$$

①

$$ii) \frac{1}{4} V_{TOT} = \frac{1}{4} \times \frac{\pi (3)^3}{27}$$

$$= \frac{\pi}{4}$$

①

when $V = \frac{\pi}{4}$

Rate of increase in h is:

$$\frac{\pi}{4} = \frac{\pi h^3}{27}$$

$$\frac{27}{4} = h^3$$

$$h = \frac{3}{4^{\frac{1}{3}}}$$

$$= \frac{3}{2^{\frac{2}{3}}}$$

①

$$\frac{dh}{dt} = \frac{9}{2\pi \left(\frac{3}{2^{\frac{2}{3}}}\right)^2}$$

$$= \frac{2^{\frac{4}{3}}}{2\pi}$$

$$= \frac{2^{\frac{1}{3}}}{\pi}$$

$$= \frac{\sqrt[3]{2}}{\pi}$$

①

b) i) $y = \log x$

$y = mx$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} = m$$

$$\Rightarrow m = \frac{1}{x}$$

at pt of intersection

$$mx = \log x$$

$$\frac{x}{x} = \log x$$

$$1 = \log x$$

$$e = x$$

$\therefore m = \frac{1}{e}$ when $y = mx$ is a tangent to $y = \log x$

ii) The range of values for which $kx = \log x$ has two distinct roots is

$$0 < k < \frac{1}{e}$$

c) $\frac{\sin 50}{\sin 0} - \frac{\cos 50}{\cos 0} = \frac{\sin 50 \cos 0 - \cos 50 \sin 0}{\sin 0 \cos 0}$

$$= \frac{\sin (50 - 0)}{\sin 0 \cos 0}$$

$$= \frac{\sin 40}{\frac{1}{2} \sin 20}$$

$$= \frac{4 \sin 20 \cos 20}{\sin 20} = 4 \cos 20$$

6) a) Prove that $2+10+24+\dots+n(3n+1)=n^2(n+1)$ for all integers $n \geq 1$,

Let $n=1$

$$\text{LHS} = 2$$

$$\text{RHS} = 1^2(1+1)$$

$$= 1 \times 2$$

$$= 2$$

$\therefore$ proposition is true for $n=1$

(1)

Assume true for $n=k$; where $k \geq 1$

$$2+10+24+\dots+k(3k+1)=k^2(k+1)$$

(1)

Prove true for $n=k+1$

$$2+10+24+\dots+k(3k+1)+(k+1)(3(k+1)+1)=(k+1)^2(k+1+1)$$

$$\text{LHS} = k^2(k+1)+(k+1)(3k+1+1) \quad \text{RHS} = (k+1)^2(k+2)$$

$$= (k+1)(k^2+3k+3+1)$$

$$= (k+1)(k^2+3k+4)$$

$$= (k+1)(k+1)(k+2)$$

$$= (k+1)^2(k+2)$$

(1)

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ proposition is true for $n=k+1$ if it is true for $n=k$.

$\therefore$ As the proposition is true for $n=1$ it is true for $n \geq 1$.

6) b) i) D: $x \in \mathbb{R}, -1 < x < 1$ (1)

ii) let $u = x$ and $v = (1 - x^2)^{\frac{1}{2}}$

$$u' = 1$$

$$v' = \frac{1}{2} (1 - x^2)^{-\frac{1}{2}} \times -2x$$

$$= -x (1 - x^2)^{-\frac{1}{2}}$$

$$y' = \frac{v u' - u v'}{v^2}$$

$$= \frac{(1 - x^2)^{\frac{1}{2}} + x^2 (1 - x^2)^{-\frac{1}{2}}}{(1 - x^2)}$$
 (1)

$$= \frac{(1 - x^2)^{\frac{1}{2}} + x^2 (1 - x^2)^{-\frac{1}{2}}}{1 - x^2} \times \frac{(1 - x^2)^{\frac{1}{2}}}{(1 - x^2)^{\frac{1}{2}}}$$

$$= \frac{1 - x^2 + x^2}{(1 - x^2) (1 - x^2)^{\frac{1}{2}}}$$
 (1)

$$= \frac{1}{(1 - x^2)^{\frac{3}{2}}}$$

iii) $m_{\text{tangent}} = \frac{1}{(1 - 0^2)^{\frac{3}{2}}}$

$$= 1$$

$$f(0) = \frac{0}{\sqrt{1 - 0^2}}$$

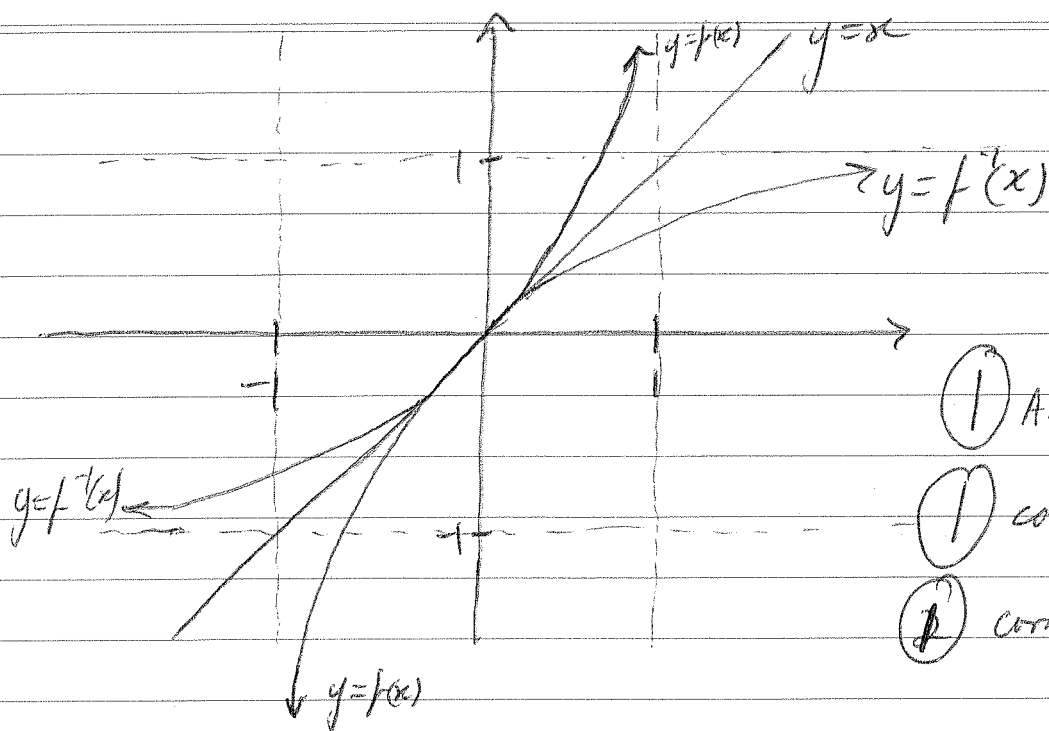
$$= 0$$

$$y - 0 = 1(x - 0)$$

eqⁿ of tangent: $y = x$

(1)

iv)



① Asymptotes

① correct $y=f(x)$

① correct $y=f^{-1}(x)$

$f(-x)$ is $-ve$

$f'(x)=1$ when $x=0$

$f(x)$ is $+ve$

v) $y=f(x)$

$$y = \frac{x}{\sqrt{1-x^2}}$$

$\therefore$ inverse is:

$$x = \frac{y}{\sqrt{1-y^2}} \quad \text{①}$$

$$x\sqrt{1-y^2} = y$$

$$x^2(1-y^2) = y^2$$

$$x^2 - x^2y^2 = y^2$$

$$y^2 = \frac{x^2}{1+x^2}$$

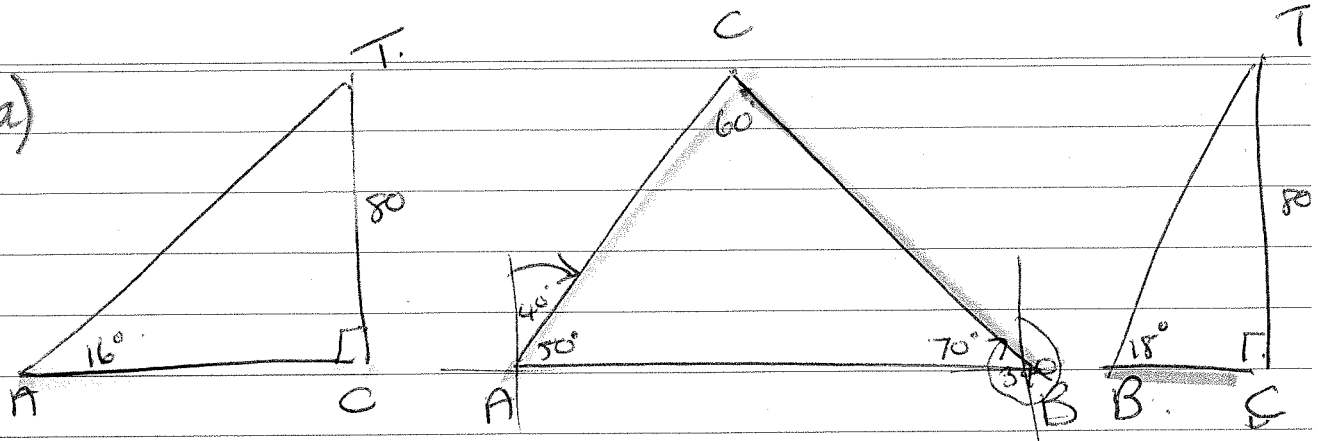
$$-y^2 - x^2y^2 = -x^2$$

$$y = \frac{x}{\sqrt{1+x^2}} \quad \text{①}$$

$$y^2 + x^2y^2 = x^2$$

$$y^2(1+x^2) = x^2 \therefore f^{-1}(x) = \frac{x}{\sqrt{1+x^2}}$$

7) a)



$$\angle ACB = 60^\circ$$

$$\angle CAB = 50^\circ$$

$$\angle CBA = 70^\circ$$

$$\therefore \text{Sum of } \angle = 180^\circ$$

In ΔACT :

$$\frac{AC}{80} = \cot 16^\circ$$

$$AC = \frac{80}{\tan 16^\circ}$$

In ΔBCT :

$$\frac{BC}{80} = \cot 18^\circ$$

$$BC = \frac{80}{\tan 18^\circ}$$

In ΔABC

By cosine rule:

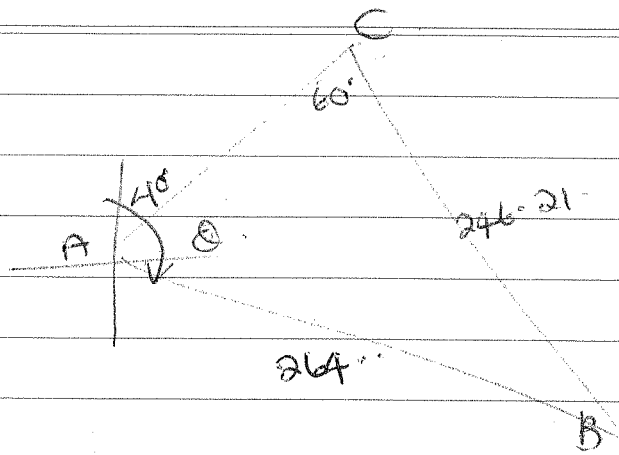
$$AB^2 = AC^2 + BC^2 - 2(AC)(BC) \cos 60^\circ$$

$$AB^2 = \left(\frac{80}{\tan 16^\circ}\right)^2 + \left(\frac{80}{\tan 18^\circ}\right)^2 - 2\left(\frac{80}{\tan 16^\circ}\right)\left(\frac{80}{\tan 18^\circ}\right) \cos 60^\circ$$

$$AB^2 = 69766$$

$$AB = 264.13$$

$$\underline{AB = 264.13} \quad (\text{to nearest m})$$



$$\frac{\sin \theta}{246.21} = \frac{\sin 60}{264}$$

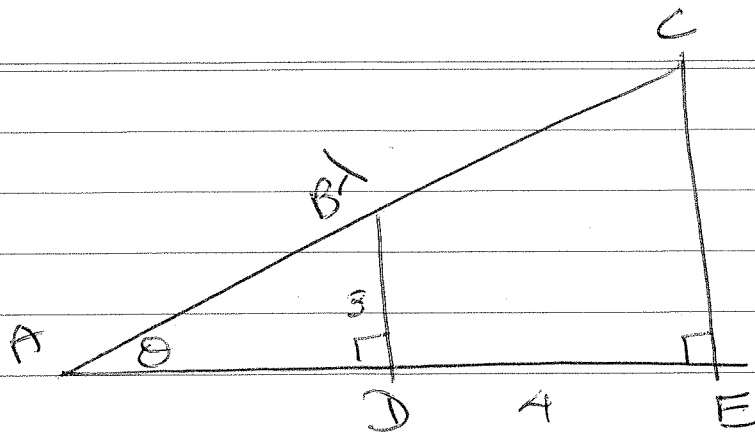
$$\sin \theta = \frac{\sin 60}{264} \times 246.21$$

$$\sin \theta = 0.507 \quad \checkmark$$

$$\theta = 53^{\circ} 50'$$

$$\begin{aligned} \therefore \text{bearing} &= 40^{\circ} + 53^{\circ} 50' \quad \checkmark \\ &= 93^{\circ} 50' \end{aligned}$$

7) b) i)



$\tan \theta = \frac{3}{AD}$ $AD = \frac{3}{\tan \theta} \checkmark$	In $\triangle ACE$: $\cos \theta = \frac{AE}{\lambda}$ $\lambda = \frac{AE}{\cos \theta}$
---	--

$$\lambda = \frac{AD + 4}{\cos \theta} \checkmark$$

$$\lambda = \frac{\frac{3}{\tan \theta} + 4}{\cos \theta} \checkmark$$

$$\lambda = \frac{3 \cos \theta}{\sin \theta} \times \frac{1}{\cos \theta} + \frac{4}{\cos \theta} \quad 2.$$

$$\lambda = \frac{3}{\sin \theta} + \frac{4}{\cos \theta} \checkmark$$

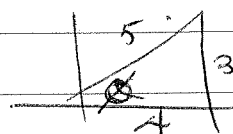
if $\lambda = 10$: $10 = \frac{3 \cos \theta + 4 \sin \theta}{\sin \theta \cos \theta}$

$$10 \sin \theta \cos \theta = 3 \cos \theta + 4 \sin \theta \quad \checkmark$$

$$5(2 \sin \theta \cos \theta) = 3 \cos \theta + 4 \sin \theta \quad \checkmark$$

$$2 \sin \theta \cos \theta = \frac{3}{5} \cos \theta + \frac{4}{5} \sin \theta \quad \checkmark$$

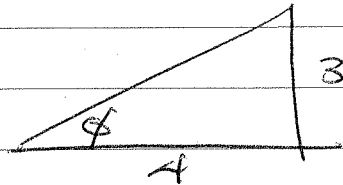
$$\begin{aligned} \sin 2\theta &= \sin \phi \cos \theta + \cos \phi \sin \theta \\ &= \sin(\theta + \phi) \quad \checkmark \end{aligned}$$



$$\text{iii)} \quad \sin 2\theta = \sin(\theta + \phi)$$

$$\therefore 2\theta = \theta + \phi \quad \checkmark \quad \text{or} \quad 2\theta = 180^\circ - (\theta + \phi) \quad \checkmark$$

$$\theta = \phi \quad \quad \quad 3\theta = 180^\circ - \phi$$



$$\phi = \tan^{-1} \frac{3}{4}$$

$$\phi = 36^\circ 52' \quad \text{or}$$

$$\therefore \theta = 36^\circ 52' \quad \checkmark \quad \text{or} \quad 3\theta = 180^\circ - 36^\circ 52'$$

$$3\theta = 143^\circ 8'$$

$$\theta = 47^\circ 43' \quad \checkmark$$
